

# Can Virtual AI Agents Improve Cognitive Assessment? A Virtual Reality Perspective

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## Abstract

We introduce SmartSage, an intelligent virtual AI agent powered by Large Language Models (LLMs) and Vision Language Models (VLMs) that enhances cognitive assessment through adaptive, embodied interactions in virtual environments. As an integral component of our Virtual Reality-based ADHD Assessment System, SmartSage serves as an interactive guide that demonstrates tasks, delivers real-time feedback, and dynamically adjusts task difficulty based on user performance. Based on the clinically validated Automated Test of Embodied Cognition (ATEC) framework, our system leverages motion tracking and haptic feedback of the Meta Quest platform to facilitate bidirectional sensorimotor interactions that boost engagement while maintaining clinical validity. By combining embodied cognition principles with virtual reality technology and AI-driven interaction, SmartSage makes cognitive assessment more accessible, engaging, and ecologically valid. In this paper, we provide an overview of the system design, present initial pilot results, and outline a roadmap for future validation studies.

## CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

## Keywords

cognitive assessment, virtual reality, executive function, embodied cognition

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## 1 Introduction

Attention Deficit Hyperactivity Disorder (ADHD) continues to be a significant social issue that impacts numerous children and adolescents [23]. The intricate features of ADHD, which include inattention, impulsivity, and sometimes hyperactivity, demand innovative approaches for precise cognitive assessment and diagnosis. ADHD frequently presents alongside other executive function (EF) challenges, which demonstrates the requirement for thorough assessment methods.

Historically, methods for assessing ADHD, like the NIH Toolbox for Cognitive Function, have typically relied on computerized tests. These evaluations gauge executive functions predominantly through tasks performed on a screen. While such instruments offer consistent metrics for cognitive abilities, they can fall short in reflecting the fluid character of attention and executive function within everyday situations where bodily movement and engagement with the surroundings are integral. Such a shortfall makes clear the importance of developing assessment techniques that can more faithfully represent how cognitive abilities manifest in natural settings.

To tackle this issue, a significant step forward has been the creation of the Automated Test of Embodied Cognition (ATEC). This system utilizes cognitively engaging physical activities to evaluate cognitive performance [2]. Differing from conventional digital tests, ATEC is grounded in Embodied Cognition (EC), a perspective holding that our thinking processes are deeply influenced by how our whole body interacts with the physical and social world around us. From this viewpoint, attention and executive function are expressed not solely via internal thought but also through the connected interplay of mind and body as individuals engage in organized physical tasks.

Extending this groundwork, Jaiswal et al. introduced the SmartFunction system [12], which led the way in transferring ATEC tasks into a virtual reality setting. While SmartFunction showed that VR-based cognitive evaluation is viable, the field still requires more interactive and tailored methods to boost participant involvement and the accuracy of evaluations, all while upholding the core tenets of embodied cognition.

This paper presents an advanced VR-based cognitive assessment system that introduces SmartSage, an interactive virtual AI agent powered by both Large Language Models (LLMs) and Vision Language Models (VLMs). SmartSage administers cognitive

assessments in virtual environments by guiding users through tasks, providing real-time feedback, and dynamically adapting to individual performance. Unlike traditional computerized assessments or previous systems that focused solely on task accuracy, our system implements a comprehensive assessment approach that evaluates both task accuracy and rhythm accuracy within physically engaging tasks - a crucial metric for ADHD assessment that aligns with embodied cognition principles. This dual-metric approach provides a more nuanced understanding of cognitive function and attention deficit patterns as they manifest through physical movement and interaction. Furthermore, our system moves beyond theoretical frameworks to provide a working prototype with planned validation through user studies. The system generates instant scores based on the performance of tasks, which can be used to assess ADHD and to develop personalized training regimens tailored to individual needs, while maintaining the fundamental connection between physical movement and cognitive function.

The main contributions of this paper are:

- Implementation of SmartSage, an interactive virtual avatar that provides real-time demonstration, instruction, and personalized feedback powered by Large Language Model and Vision Language Model.
- Development of a comprehensive assessment system that evaluates both task accuracy and rhythm accuracy in Attention Deficit assessment through embodied cognition principles.
- Creation of a working prototype with planned user validation studies that bridges the gap between traditional computerized evaluation and physical task-based evaluation.

## 2 Related Work

Traditional methods for cognitive assessment and rehabilitation of attention primarily rely on pencil-and-paper techniques, motor reaction time tasks, and computerized testing programs [9, 15, 21]. The Stroop test [22] stands as one of the most recognized neuropsychological assessments for attentional dysfunction, requiring participants to name the color of printed words while suppressing the automatic reading response. Similarly, the flanker task [7] measures interference control by having participants respond to a target arrow while ignoring flanking arrows pointing in opposite directions [8]. The NIH Toolbox Cognition Battery represents another standardized approach, providing computerized tests to measure executive functions through screen-based tasks [26]. However, these traditional neuropsychological tests assess cognitive constructs in isolation, without considering real-world context, resulting in limited predictive potential for everyday performance [19]. Studies have demonstrated that the relationship between performance on paper and pencil EF tasks and performance in activities of daily living is notably weak [4].

The concept of embodied cognition proposes that cognitive processes are fundamentally shaped by physical interactions with the environment [2]. The Automated Test of Embodied Cognition (ATEC) implements this principle through cognitively demanding physical tasks that measure executive function in action. ATEC uses motion capture of physical movements and applies computer vision algorithms to score performance and assess attention [6, 29].

This approach has shown promising results in evaluating executive function through natural physical movements rather than abstract computerized tasks. Unlike traditional assessments, ATEC considers the intricate relationship between body movement and cognitive processes during structured activities, offering a more comprehensive understanding of cognitive function [18]. Research has shown that ATEC serves as a better predictor of children's functioning compared to conventional testing methods [2].

Virtual Reality (VR) has emerged as a powerful tool for cognitive assessment, offering immersive environments that can simulate real-world scenarios while maintaining experimental control. Previous studies have demonstrated successful replication of clinical assessment tasks in VR [10, 17]. VR has proven effective for evaluating various executive function-related abilities, including task-switching, working memory [13], and spatial memory [14]. Recent work by Williams et al. [27] has developed a toolkit for creating virtual interactions that replicate traditional cognitive tests while maintaining measurement validity. Research has shown that participants exhibit more natural behaviors when immersed in virtual environments that simulate daily activities [3], and performance in virtual tasks correlates significantly with real-world performance [11, 20].

Recent advances in artificial intelligence, particularly Large Language Models (LLMs) and machine learning, have opened new possibilities in cognitive assessment. Researchers have utilized machine learning algorithms to analyze movement patterns, predict cognitive decline, and provide personalized feedback in assessment tasks [28]. Studies have shown success in using AI to classify between individuals with cognitive impairments and healthy participants [24]. The integration of AI with virtual reality environments has enabled more sophisticated analysis of user behavior and cognitive patterns, allowing for real-time adaptation of assessment tasks and personalized intervention strategies [1]. This raises an intriguing possibility: agent-based EF assessment, where an AI agent within a virtual environment conducts and adapts the evaluation much like a human examiner or coach. An AI agent could present tasks, guide the user, observe performance (potentially through multimodal inputs like actions and verbal responses), and adjust the difficulty or strategy of the assessment on the fly. Such a system could personalize the test to each individual's profile, for example, selecting different toolsets or task variations for users with different levels of ability or needs. This personalization could make assessments more engaging and more accurate by targeting the right level of challenge for each user [5].

Our work builds upon these foundations by combining the ecological validity of ATEC's embodied cognition principles with the immersive capabilities of VR technology. By introducing SmartSage, an LLM-powered virtual avatar, we enhance the assessment experience through interactive guidance and personalized feedback. This integration of embodied cognition, VR technology, and AI-driven interaction represents a novel approach to cognitive assessment that maintains the benefits of physical movement-based evaluation while adding the standardization and accessibility of virtual environments.

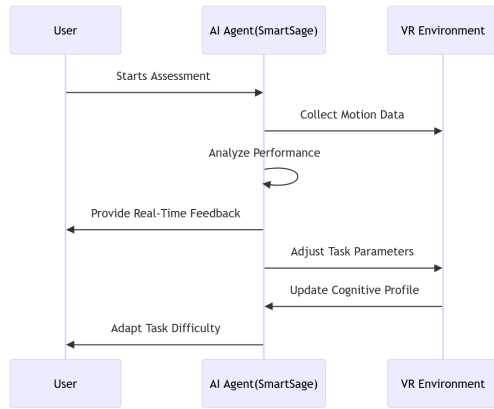


Figure 1: Overview of the interaction between the User, SmartSage, and the VR system.

### 3 System Design

This section details the architecture and implementation of our new virtual reality system that transforms traditional ATEC cognitive assessments into an immersive VR experience. Built on the Meta Quest platform and Unity game engine, the system creates a controlled virtual environment where users perform validated assessment tasks, Pass the Ball to the Beat (PBB) and Cross Your Body (CYB) - under the guidance of SmartSage, an adaptive virtual avatar [16, 25]. Through this integration of VR technology with embodied cognition principles, we aim to provide standardized, engaging cognitive assessments that closely mirror real-world interactions.

#### 3.1 Core Assessment Tasks

Our system implements two key tasks from the Automated Test of Embodied Cognition (ATEC), a clinically validated assessment system that evaluates executive function through physical movement. By precisely measuring movement patterns and response timing, ATEC enables a comprehensive assessment of attention disorders and broader cognitive functions. The system incorporates two primary ATEC tasks:

**Pass the Ball to the Beat (PBB) task:** This task assesses sustained attention and response inhibition through coordinated rhythmic movements. Participants interact with a virtual ball following three commands:

- "Green Light": Pass the ball between hands
- "Yellow Light": Move the ball vertically
- "Red Light": Hold the ball still

Commands are delivered through both audio cues and visual traffic light signals, enabling assessment across different sensory modalities. Task difficulty increases progressively through adjustments in rhythm, speed, and command sequencing.

**Cross Your Body (CYB) task:** This task evaluates multiple cognitive domains through a series of synchronized body-crossing movements. The sequence involves:

- Touching opposite body parts (e.g., left hand to right ear)
- Following rhythmic audio cues.

- Progressing through different body targets (ears, knees, hips, shoulders)
- Responding to inverse commands in final rounds (e.g., touching knees when "ears" is called)

The task challenges participants' sustained attention, working memory, and response inhibition while requiring precise bilateral coordination. Movements across the midline and inverse commands in later stages significantly increase cognitive load, providing deeper insight into the capabilities of executive function.

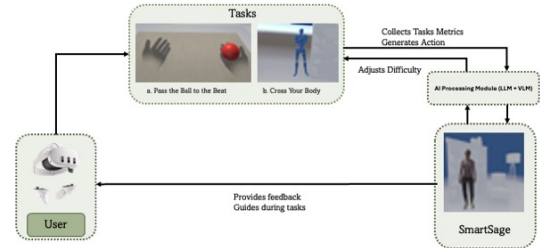


Figure 2: Main components of the system.

#### 3.2 SmartSage Virtual AI Agent

SmartSage serves as an intelligent virtual AI agent in our VR-based cognitive assessment system. Powered by both LLMs and VLMs, SmartSage not only delivers verbal commands and visual demonstrations but also interprets visual and contextual cues from the user's interactions. This capacity for multimodal understanding permits SmartSage to fluidly adjust the challenge level of tasks, deliver individualized support, and uphold the assessment's appeal and clinical integrity. To foster authentic, two-way exchanges, the avatar utilizes the Meta Quest's systems for motion tracking and haptic response. By means of visual demonstrations, spoken directives, and physical gestures, SmartSage shepherds participants through progressively demanding assessment activities, all while preserving participant interest and uniform testing conditions.

The avatar's behavior adapts dynamically based on user performance metrics, including movement accuracy, response timing, and task completion rates. When users struggle, SmartSage provides additional instruction or simplifies task complexity; conversely, it increases difficulty for users demonstrating mastery. This adaptive interaction is driven by machine learning algorithms that analyze real-time movement data captured through VR controllers, enabling personalized assessment experiences while maintaining clinical validity. Figure 1 shows the interaction between the user, SmartSage, and the VR system.

#### 3.3 System Architecture

Our system consists of three main components: (1) the user, who interacts with the environment via a Meta Quest headset; (2) the tasks, which include "Pass the Ball to the Beat" and "Cross Your Body"; and (3) SmartSage, an AI-driven virtual guide. As the user

performs each task, the Meta Quest headset and controllers capture sensorimotor data such as hand position, movement precision, and response timing. Figure 2 shows the main components of the system and the interaction between them.

This data streams continuously to the AI Processing Module, which employs compact Large Language Models (LLMs) and multimodal models to interpret user actions in real time. The module integrates lightweight LLMs like Phi 3 for natural language processing and task adaptation. For multimodal processing, PaliGemma 2 analyzes both visual data and text to decode user movements and contextual cues within the virtual environment. By combining textual and visual insights, the module dynamically adjusts task difficulty and delivers tailored feedback based on individual performance.

SmartSage then relays personalized guidance, demonstrations, and corrective cues to the user, closing the feedback loop. This iterative cycle of data capture, AI-driven analysis, and adaptive feedback enables an immersive, embodied cognitive assessment experience—one that tailors itself to each individual’s skill level and real-time performance.

4 User Study and Preliminary Results

We conducted a pilot user study with eight participants to evaluate the VR-based cognitive assessment system featuring SmartSage. The participants were aged between 25 and 32, and the group consisted of five female and three male participants. Initially, participants were briefed on the fundamentals of cognitive assessment, the principles of embodied cognition, and the ATEC framework. They then experienced demonstrations of both systems: the traditional ATEC system with Eliza and the VR-based immersive assessment with SmartSage. Eliza— a recorded video of a human female guide—provided step-by-step instructions and task demonstrations in the traditional ATEC. After engaging with both systems, the participants completed a comprehensive survey that compared various aspects, including task comprehension, performance confidence, user engagement, and quality of feedback. Notably, 5 participants reported no prior experience with VR, and 3 participants reported limited to moderate familiarity with VR. Results from the comparative evaluation of the cognitive assessment systems revealed several compelling trends in user preferences and system effectiveness. Although 5 out of 8 participants favored ATEC with Eliza for initial task understanding and performance, SmartSage demonstrated clear advantages in other critical areas. For example, SmartSage received unanimous positive ratings for feedback quality, and most participants rated the VR system higher in terms of engagement and confidence gained.

Participants rated the SmartSage experience favorably in multiple dimensions. The clarity of the instructions received strong endorsement, with 7 out of 8 participants agreeing or strongly agreeing that the SmartSage instructions were clear and easy to follow. Moreover, 6 out of 8 participants felt that the system’s responses were natural and appropriately contextualized. Although interactions with VR controllers were generally positive, there was some variability in comfort levels, which highlighted an area for further refinement. Figure 3 shows a summary of the user experience of the VR-based assessment system powered by SmartSage.

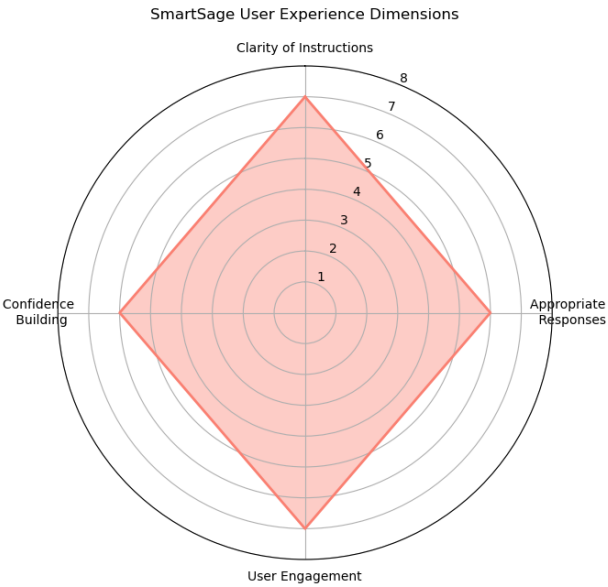


Figure 3: User experience for VR-based system powered by SmartSage.

Regarding future preferences, Figure 4 indicates that the majority of

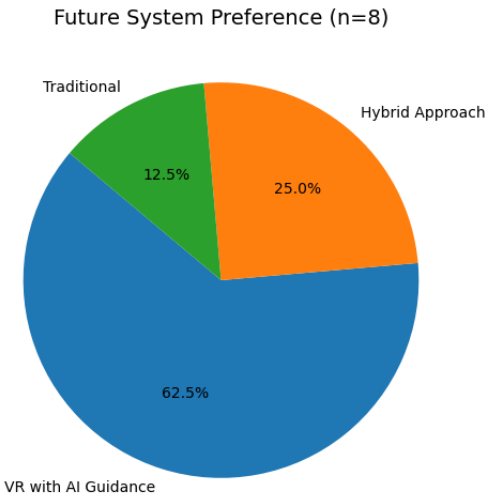


Figure 4: Future preferences for assessment methods.

participants expressed a preference for VR with AI guidance in subsequent cognitive assessments. Two participants suggested a hybrid approach that combines both systems, and only one participant

preferred the traditional video-guided method exclusively. This distribution underscores a strong inclination towards innovative assessment methods while also recognizing the value of established approaches.

Qualitative feedback further emphasized the advantages of the SmartSage system. Many participants appreciated the adaptive nature of AI guidance and the immersive quality of the virtual reality environment. However, some feedback indicated that the initial task explanation phase could be improved, as the straightforward approach of the traditional system was perceived as more effective during the early stages of the assessment.

## 5 Conclusion

This paper introduces SmartSage, an innovative virtual AI agent powered by advanced LLM and VLM technologies that directs the cognitive assessment process within the SmartFunction VR assessment system. Leveraging its bidirectional sensorimotor interaction capabilities and validated tasks, such as Pass the Ball (PBB) and Cross Your Body (CYB), SmartSage creates an adaptive and engaging assessment environment that captures rich diagnostic data while maintaining high user engagement. By integrating embodied cognition principles with cutting-edge AI, SmartSage offers a versatile tool suitable for a wide range of applications, from educational institutions to healthcare providers, by delivering standardized testing conditions that can be tailored to individual user needs. In the future, our research will focus on implementing personalization algorithms for detailed cognitive profiling and conducting extensive user studies to validate the system's effectiveness across diverse populations. These advancements will further enhance SmartSage's capability to provide meaningful cognitive measures while preserving its accessibility and ease of use.

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